Repair Bias in Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Recall and choose your starting point

> Record your starting explanation for Repair Bias in Data before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sampling and Representation

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Noise and Outliers

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Repair Bias in Data trace and record the first step you can explain without help.

Use W2: Combine counts for **Selection bias** and coverage. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Repair Bias in Data explanation against the supplied facts before discussing it.

Use the worked case for **Selection bias** and coverage. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Audit independently

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Repair Bias in Data target to a trace and a justified conclusion.

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Repair Bias in Data case and show the deciding evidence.

Training has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

Propose repair and evaluation.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A dataset has 18 adventure readers and 2 poetry readers. A recommender ignores poetry preferences. What would you inspect and test after a repair?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 20 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Repair Bias in Data adds to Transparent Book Recommender and keep the evidence.

Change feature coverage or weights only after locating which reader profiles are underserved.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Locate undercoverage

> Locate undercoverage. Apply the supplied Repair Bias in Data facts and keep your reasoning.

A morning-only survey claims to represent all library visitors. Which visitors may be missed?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Combine counts

> Combine counts. Apply the supplied Repair Bias in Data facts and keep your reasoning.

Group A has 3 Yes of 4 and B has 2 Yes of 6. Find the combined proportion.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair the collection

Repair the collection. Apply the supplied Repair Bias in Data facts and keep your reasoning.

A form is shared only in the robotics club. How can the **process** better cover the grade?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Audit independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Audit independently. Apply the supplied Repair Bias in Data facts and keep your reasoning.

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

My reasoning and evidence

W5 Avoid causal claims

> Avoid causal claims. Apply the supplied Repair Bias in Data facts and keep your reasoning.

Club members favor coding more often. Does club membership necessarily cause that preference?

My reasoning and evidence

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Repair Bias in Data

> A fictional school polls learners about adding comic recommendations. Explain the labelled step before checking the changed condition.

A fictional school polls learners about adding comic recommendations.

Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15.

The full grade has 30 club members and 90 others. These respondents were volunteers, not random selections.

AI ACADEMY / AI-2/20

Volunteer group comparison

	Yes responses	Respondents	Grade members
Club	12	15	30
Others	3	15	90

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/20

Repair Bias in Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Group rates

Club:12/15=80%; others:3/15=20%.

2. Respondent total

15 Yes out of 30 respondents =50%.

3. Check representation

Respondents split 50/50, but the grade is 25% club and 75% other.

4. Repair collection

Plan invitations that fairly cover the whole grade, document selection and report nonresponse by group.

CHANGE ONE THING

Recruit more club volunteers only.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional school polls learners about adding comic recommendations.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group.

1. Group rates

2. Respondent total

3. Repair collection

4. Check representation

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair collection Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group.

The combined respondent result gives club members more influence than their share of the grade.

A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Recruit more club volunteers only. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Sample size increases without fixing missing representation of other learners.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Sample size increases without fixing missing representation of other learners. The added respondents come from the already overrepresented group.

Repair a missing building block

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can **80%** response be reported as **80%** of the whole grade responding?. Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?.

Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

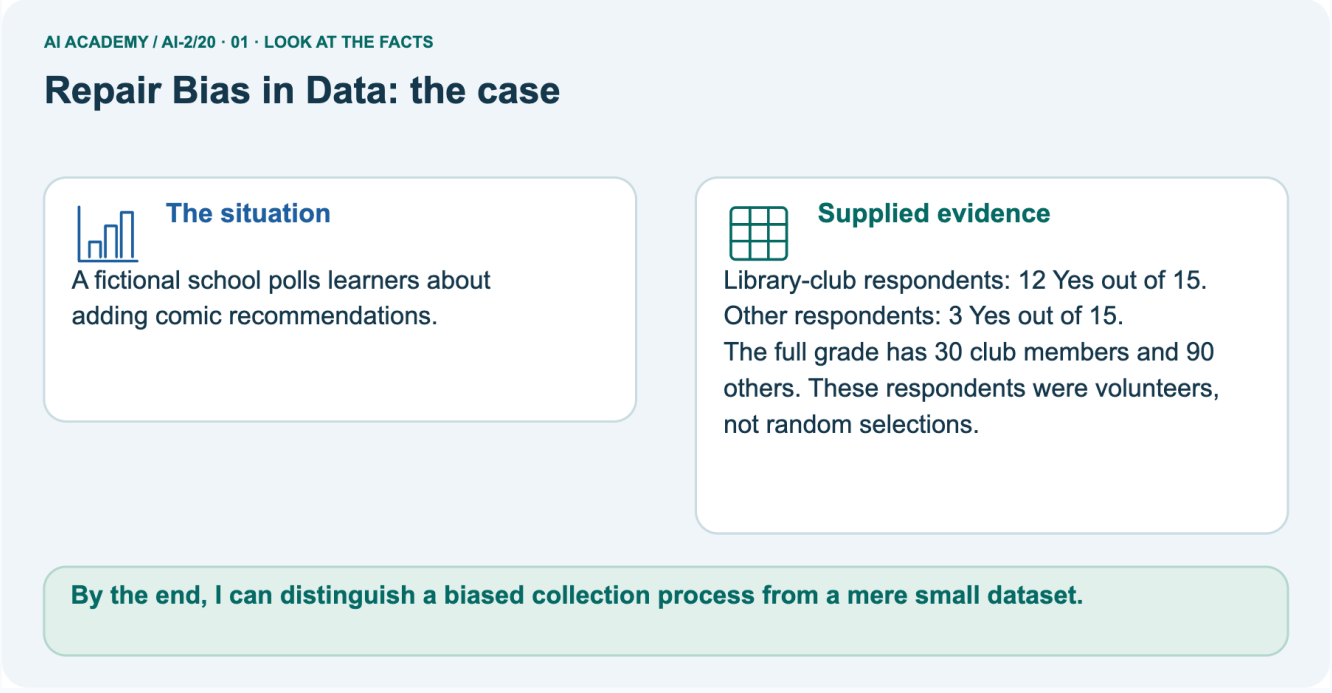
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Repair Bias in Data picture from its labels, then record a relevant limit.

01. Repair Bias in Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

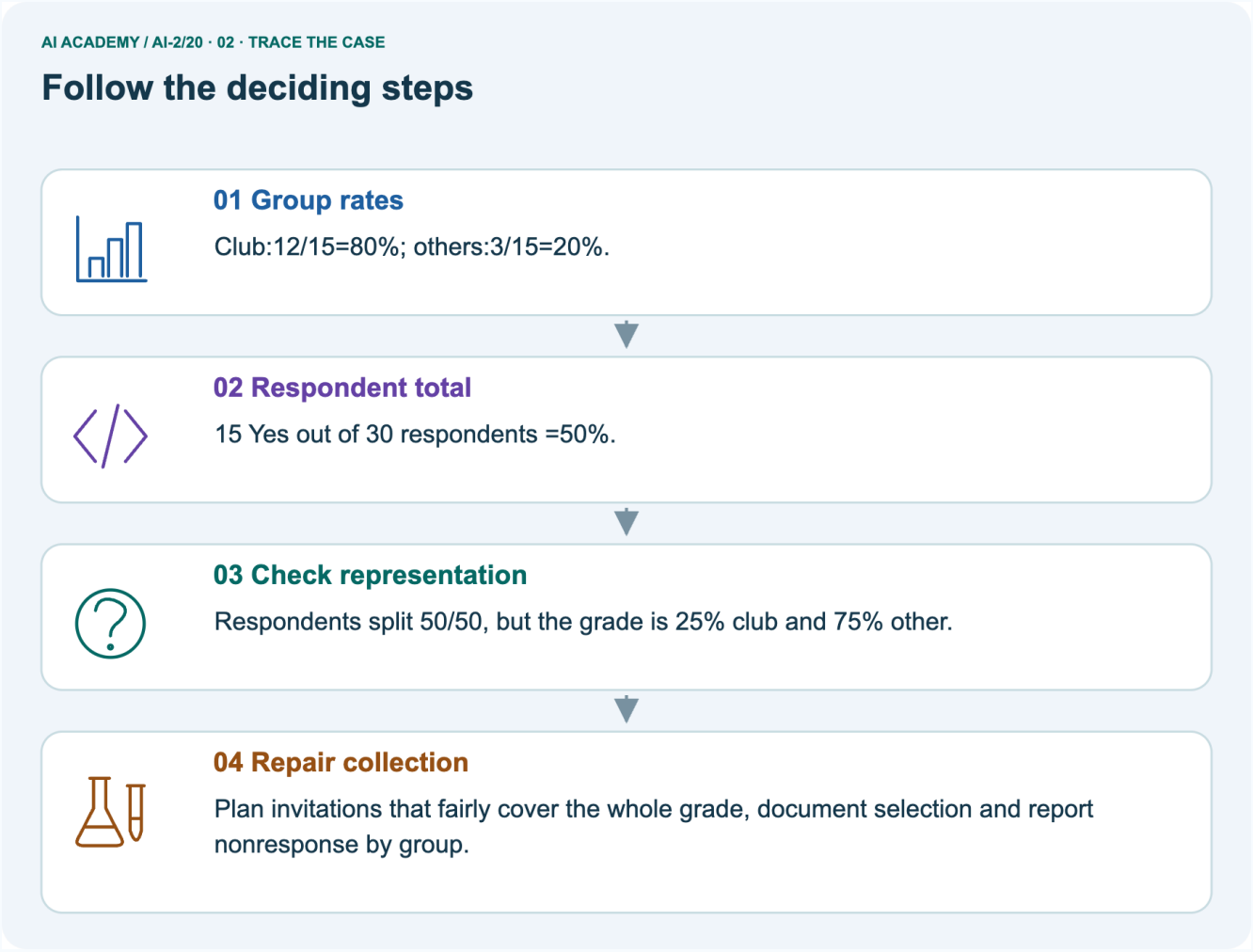
A fictional school polls learners about adding comic recommendations. Supplied facts: Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15. The full grade has 30 club members and 90 others.

These respondents were volunteers, not random selections.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



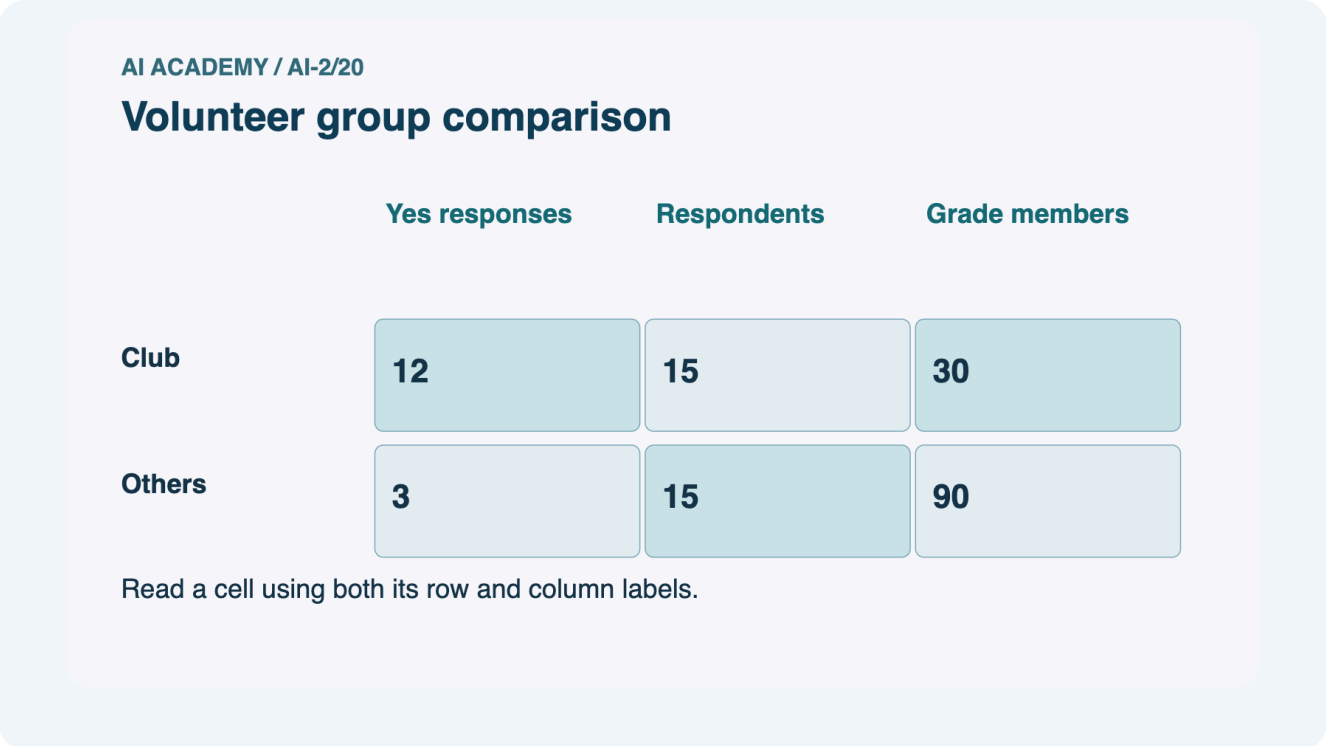
Follow the labelled picture with the matching explanation. Open the vector diagram

Group rates: Club:12/15=**80%**; others:3/15=20%. → Respondent total: 15 Yes out of 30 respondents =50%. → Check representation: Respondents split 50/50, but the grade is 25% club and 75% other. → Repair collection: Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Volunteer group comparison



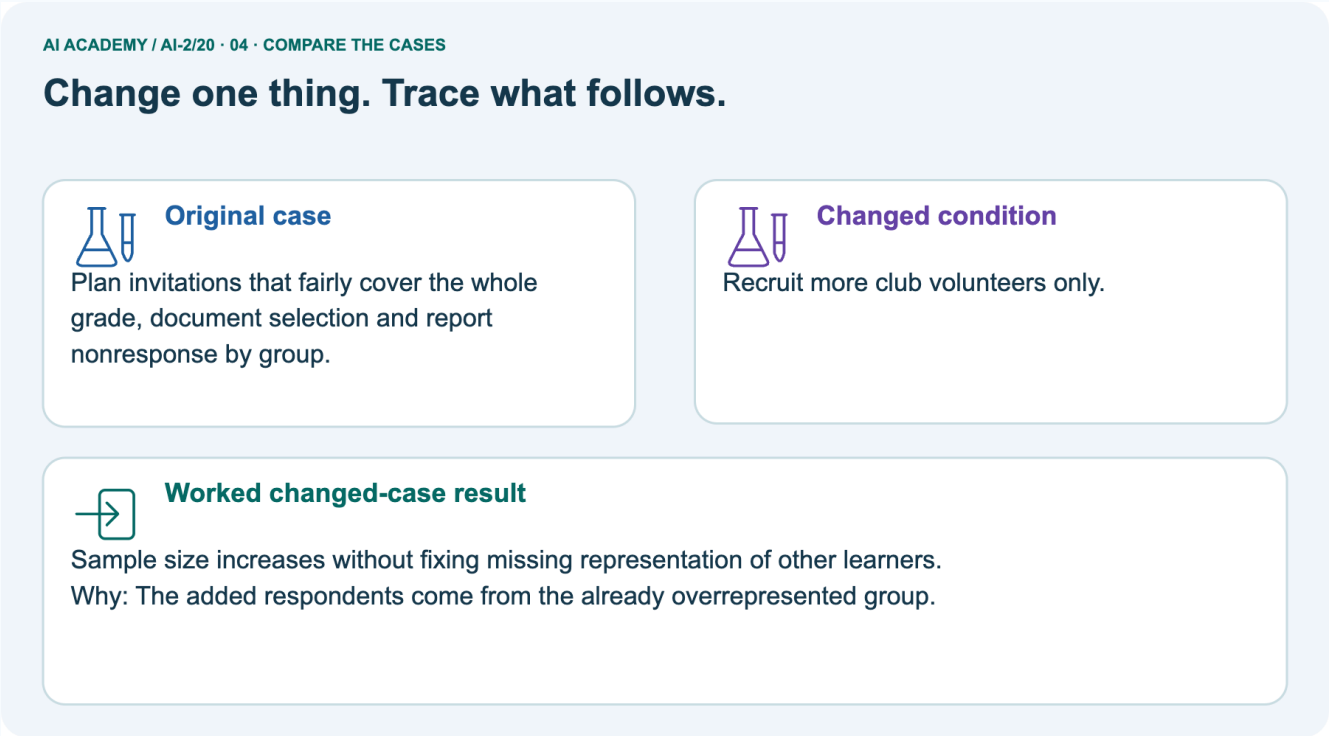
Follow the labelled picture with the matching explanation. Open the vector diagram

Volunteer group comparison. Club: Yes responses = 12, Respondents = 15, Grade members = 30. Others: Yes responses = 3, Respondents = 15, Grade members = 90.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

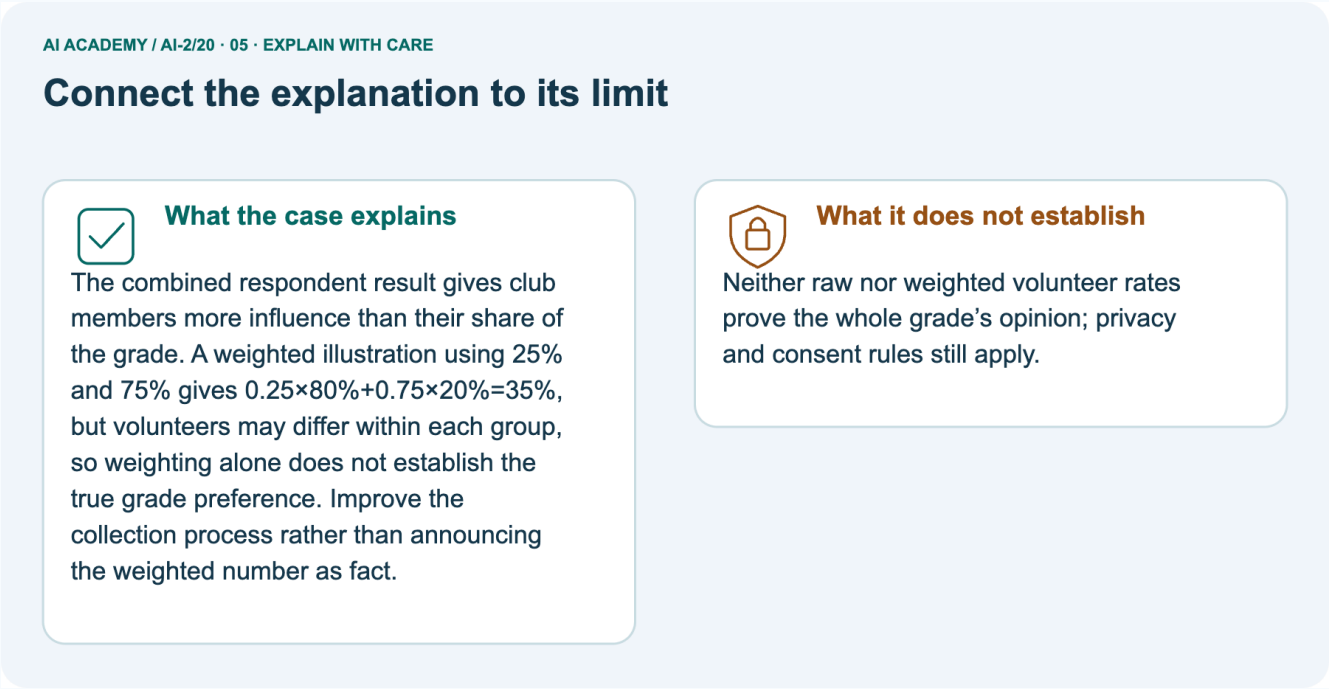
Original case: Plan invitations that fairly cover the whole grade, document selection and report **nonresponse** by group. Changed condition: Recruit more club volunteers only.

Expected result: Sample size increases without fixing missing representation of other learners. Reason: The added respondents come from the already overrepresented group.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The combined respondent result gives club members more influence than their share of the grade.

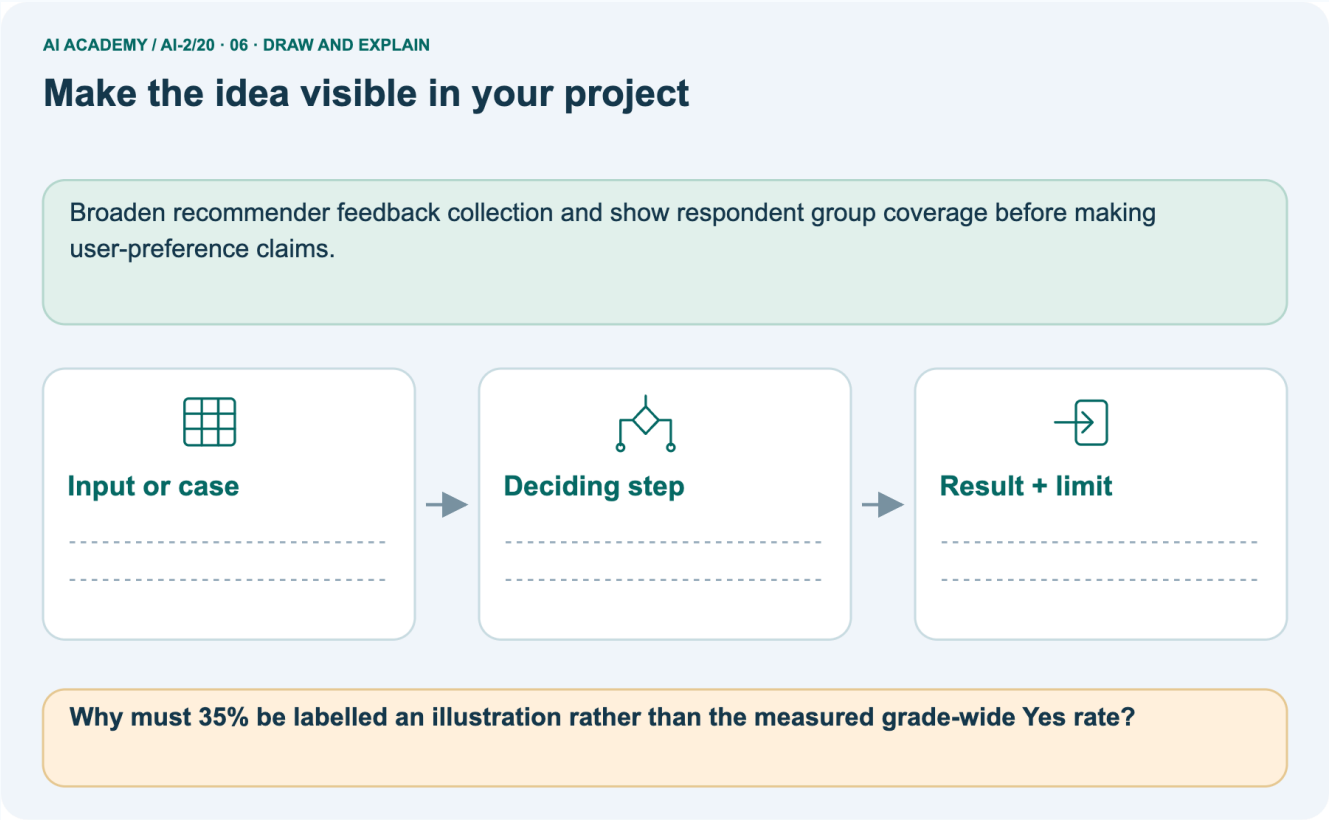
A weighted illustration using 25% and 75% gives 0.25×**80%**+0.75×20%=35%, but volunteers may differ within each group, so weighting alone does not establish the true grade preference.

Improve the collection **process** rather than announcing the weighted number as fact. Limit: Neither raw nor weighted volunteer rates prove the whole grade's opinion; privacy and consent rules still apply.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims. Draw your **input**, deciding step, result and limit.

Supported question: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.